Feature Description 544/221 04-LZA 701 6014/1 Uen AB

Uplink-Triggered VoLTE Mobility

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1 Uplink-Triggered VoLTE Mobility Overview

> This feature introduces an uplink-triggered VoLTE mobility procedure for UEs with voice bearers that is performed when fast degradation of the uplink radio conditions occurs.

Feature Name	Uplink-Triggered VoLTE Mobility
Feature Identity	FAJ 121 5433
Value Package Name	VoLTE Performance
Value Package Identity	FAJ 801 0444
Node Type	Baseband Radio Node
Access Type	LTE
Licensing	License-controlled feature. One license is required for each node.

Summary

UEs with a voice bearer can lose uplink voice packets if the uplink coverage is insufficient. This feature detects the fast degradation of uplink conditions by monitoring HARQ statistics. If uplink conditions are degraded, the feature performs intra-LTE inter-frequency measurement-based handover or blind handover to move the UE to a cell with better coverage.

The feature provides the following benefits:

- Packet loss is avoided or reduced.
- VoLTE quality is improved.

Additional Information

For more information and related topics, see Downlink-Triggered VoLTE Mobility.

For information on the preferred state and parameter settings of this feature, see RAN Parameter Recommendations Lists.

2 Dependencies of Uplink-Triggered VoLTE Mobility

> The feature improves the operation of other mobility-related features during fast degradation of uplink radio conditions.

Table 1 Feature Dependencies

Feature	Relationship	Description
Coverage-Triggered Inter-Frequency Handover (FAJ 121 0877)	Prerequisite	The Coverage-Triggered Inter-Frequency Handover feature must be activated for the Uplink-Triggered VoLTE Mobility feature to be operational.
Intra-LTE Handover (FAJ 121 0489)	Prerequisite	The Intra-LTE Handover feature must be activated for the Uplink-Triggered VoLTE Mobility feature to be operational. This is a basic feature that does not require manual activation and cannot be deactivated.
Downlink-Triggered VoLTE Mobility (FAJ 121 5686)	Related	If both features are enabled, the feature that first indicates packet loss triggers the handover. The packet loss later indicated by either of the features is discarded. This means handover is triggered only once in a single cell during a voice call. This can be adapted to most scenarios to achieve the optimal combination gain.
Intra-LTE Inter-Mode Handover (FAJ 121 3042)	Related	A handover that is triggered by the Uplink-Triggered VoLTE Mobility feature can be performed between LTE TDD and LTE FDD neighbor cells if the Intra-LTE Inter-Mode Handover feature is also activated.
Automated Neighbor Relations (FAJ 121 0497)	Related	If a measurement configured by the Uplink-Triggered VoLTE Mobility feature reports an unknown cell during handover, the Automated Neighbor Relations feature is not triggered. If there is a high rate of failed handovers, the Living with PCI Conflicts function can improve mobility KPIs, but it can also have a negative impact on voice call quality during handover to problematic cells. If the preferred cell selection in PCI conflicts function is enabled, handover success rate can improve in cell relations with PCI confusion.
Mission-Critical Push-to-Talk (FAJ 121 4929)	Related	If the Uplink-Triggered VoLTE Mobility and the Mission-Critical Push-to-Talk features are activated together, uplink-triggered VoLTE mobility can also be performed for a UE with a PTT voice bearer.
Basic Intelligent Connectivity (FAJ 121 4843)	Related	Uplink-triggered VoLTE mobility can be performed for a VoLTE UE with an established EN-DC connection from an EN-DC anchor cell to a non-EN-DC anchor cell. UE throughput can be decreased.
Mobility Control at Poor Coverage (FAJ 121 3013)	Related	If both features are enabled, the UE can be transferred to the cell during handover with the highest priority frequency layer instead of the frequency layer selected by the Mobility Control at Poor Coverage feature.
Service or Priority-Triggered Inter-Frequency Handover (FAJ 121 3087)	Related	The Uplink-Triggered VoLTE Mobility handovers are disabled if all of the following conditions are fulfilled: – The Service or Priority-Triggered Inter-Frequency Handover measurements are configured in the UE. – The value of the `EUtranCellFDD.servOrPrioTriggeredIFHo` or the

Feature	Relationship	Description
		EUtranCellTDD.servOrPrioTriggeredIFHo attribute is set to QCI. – The value of the EUtranCellFDD.servOrPrioIFHoSetupBearer or the EUtranCellTDD.servOrPrioIFHoSetupBearer attribute is set to FALSE.
Subscriber Triggered Mobility (FAJ 121 1788)	Related	The RAT and Frequency Priority Modification (RFPM) function of the Subscriber Triggered Mobility and the Uplink-Triggered VoLTE Mobility features can be enabled at the same time. Then, the value of the EUtranFreqRelation.voiceprio attribute is overridden by the corresponding parameter from the RATFreqPrio or the UeProfile MO class.
Multiple Frequency Band Indicators (FAJ 121 3054)	Related	If the Multiple Frequency Band Indicators feature is supported but the UE does not support the primary frequency band, a check for additional frequency bands towards UE capabilities is done. If the UE supports any of the additional frequency bands, the handover is initiated if all other legacy checks are fulfilled.
Device-Based Frequency Filtering (FAJ 121 5154)	Related	The Device-Based Frequency Filtering feature is used to configure a list of unsupported frequencies for specific UEs. These frequencies are excluded from UE configuration messages sent by the eNodeB. Unsupported frequencies are configured using the UnsupportedFreqProfile.unsupportedFreqList attribute.
Uplink Carrier Aggregation (FAJ 121 4425)	Related	If the Uplink Carrier Aggregation and the Uplink-Triggered VoLTE Mobility features are activated simultaneously, the HARQ responses from the SCell can also be counted. In this case, potential packet loss detection is based on the HARQ response from all cells that contribute to uplink transmission, and the handover triggering can be affected. It is recommended to configure the thresholds for potential packet loss to the same percentage by setting the EUtranCellFDD.ulVolteCovMobThr or the EUtranCellTDD.ulVolteCovMobThr attribute to the same value in the PCell and the SCell. If the values are different, the KPI results based on parameter settings are not comparable.
VoLTE Optimized Carrier Aggregation (FAJ 121 4884)	Related	If the VoLTE Optimized Carrier Aggregation and the Uplink Carrier Aggregation features are both enabled, the Uplink Carrier Aggregation feature only has effect on the HARQ response counts when the EUtranCellFDD.sCellHandlingAtVolteCall or the EUtranCellTDD.sCellHandlingAtVolteCall attribute is set to KEEP_SCELLS. Otherwise, the VoLTE Optimized Carrier aggregation feature has no impact on the Uplink-Triggered VoLTE Mobility feature.

Hardware

No specific hardware requirements.

Limitations

NB-IoT UEs and Category M1 UEs are not supported.

Network Requirements

No specific network requirements.

Related concepts

Engineering Guidelines for Uplink-Triggered VoLTE Mobility

3 Feature Operation of Uplink-Triggered VoLTE Mobility

> The feature introduces uplink-triggered VoLTE mobility for UEs with voice bearers if fast degradation of uplink radio conditions occurs.

The feature is enabled by setting the `EUtranCellFDD.ulVolteCovMobDetect` or the `EUtranCellTDD.ulVolteCovMobDetect` attribute to one of the following values:

- `BLINDHO` for blind handover
- `MEASBASEDHO` for measurement-based handover

Uplink-Triggered VoLTE Mobility is only applicable for UEs with a VoLTE or a Mission-Critical Push-To-Talk voice bearer.

3.1 Blind Handover

> During blind handover, the coverage of the target cell is assumed, and the user equipment is ordered to commence mobility without any measurements on target cells.

Blind handover is chosen by setting the `EUtranCellFDD.ulVolteCovMobDetect` or the `EUtranCellTDD.ulVolteCovMobDetect` attribute to `BLINDHO`. In this case, the `EUtranCellFDD.covTriggerdBlindHoAllowed` or the `EUtranCellTDD.covTriggerdBlindHoAllowed` attribute must be set to `true` in the serving cell.

Target Cell Selection for Uplink-Triggered VoLTE Mobility with Blind Handover

When the feature is enabled, several additional attributes need to be set for the process to be successful. The following configuration is needed for target cell selection in the candidate frequency:

- The `EUtranFreqRelation.voicePrio` attribute is configured for each frequency relation.
  Note: If multiple frequencies have the same highest priority value, all those frequencies are eligible for random selection for blind handover.

  For example, if the `EUtranFreqRelation.voicePrio` attribute is set to 6 for four frequencies, and all the other frequencies have lower values, all four frequencies with value 6 are eligible for random selection for blind handover.
- The `EUtranFreqRelation.mobilityAction` is set to `HANDOVER`.

The target cell is selected from the candidate frequency set. The following configuration is needed for the candidate cell:

- The `EUtranCellRelation.isHoAllowed` is set to `true`.
- The `EUtranCellRelation.mobilityStatus` is set to `true`.
- The `EUtranCellRelation.coverageIndicator` is set to `COVERS`.

Operational Flow

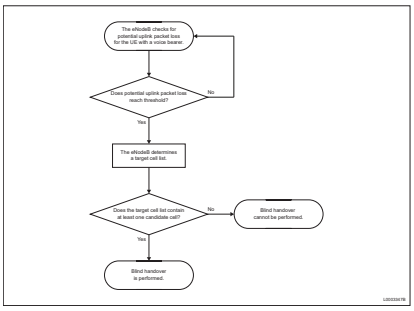

Figure 1 Uplink-Triggered VoLTE Mobility with Blind Handover

1. The potential uplink packet loss threshold can be set with the `EUtranCellFDD.ulVolteCovMobThr` or the `EUtranCellTDD.ulVolteCovMobThr` attribute.
2. The eNodeB monitors the number of uplink DTX and HARQ NACK responses since the last HARQ ACK response, then compares it with the configured threshold. If the number of responses is higher than the threshold, potential uplink packet loss is detected.
3. The eNodeB determines a set of frequency candidates based on the `EUtranFreqRelation.voicePrio` attribute values, and selects the target cell in the target frequencies. The selection is based on the configuration requirements for coverage relations specified in Target Cell Selection for Uplink-Triggered VoLTE Mobility with Blind Handover.

   If the target cell list is empty, no target cell can be selected. The handover cannot be performed, and the procedure stops.
4. The eNodeB performs blind handover to the target cell.

3.2 Measurement-Based Handover

> During measurement-based handover, the user equipment is ordered to measure the signal strength or quality of the target cell, followed by an order to handover if the measurement result meets certain criteria.

Target Cell Selection for Uplink-Triggered VoLTE Mobility with Measurement-Based Handover

When the feature is enabled, several additional attributes need to be set for the process to be successful. The following configuration is needed for target cell selection in the candidate frequency layer:

- The `EUtranFreqRelation.voicePrio` attribute is configured for each frequency relation.

  Note: If more than one frequency layer has the same highest priority value, all those frequencies are selected as candidates for handover.

  For example, if the `EUtranFreqRelation.voicePrio` attribute is set to 6 for four frequencies, and all the other frequencies have lower values, all four frequencies with value 6 are chosen as candidates for handover.

- The `EUtranFreqRelation.mobilityAction` is set to HANDOVER.

The target cell is chosen from the measurement report. The target cell in the measurement report must meet the following configuration:

- The value of the `EUtranCellRelation.isHoAllowed` attribute is `true` for at least one cell of the frequency.
- The value of the `EUtranCellRelation.mobilityStatus` attribute is set to `true` for at least one cell of the frequency.

Operational Flow

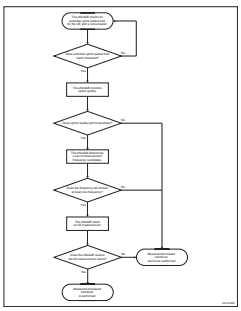

Figure 2 Uplink-Triggered VoLTE Mobility with Measurement-Based Handover

1. The potential uplink packet loss threshold can be set with the `EUtranCellFDD.ulVolteCovMobThr` or the `EUtranCellTDD.ulVolteCovMobThr` attribute.
2. The eNodeB monitors the number of uplink DTX and HARQ NACK responses since the last HARQ ACK response, then compares it with the configured threshold. If the number of responses is higher than the threshold, potential uplink packet loss is detected.
3. The eNodeB continues monitoring the link quality by checking the uplink DTX and HARQ responses. If the uplink quality recovers, or it degrades and a large number of DTX and NACK responses is detected, the handover cannot be performed and the procedure stops.
4. The eNodeB determines a set of frequency candidates based on related parameters specified in Target Cell Selection for Uplink-Triggered VoLTE Mobility with Measurement-Based Handover.

   If the frequency set is empty, no target frequency can be selected. The handover cannot be performed and the procedure stops.
5. The eNodeB starts an A5 measurement. By setting the `UeMeasControl.a5TimerUlVolteCovMob` attribute, the operator can define how long the eNodeB waits for the measurement report.
6. If the eNodeB receives the A5 measurement report, it performs measurement-based handover to the target cell.

   If the eNodeB does not receive the A5 measurement report before the timer expires, the measurement-based handover cannot be performed. The procedure also stops if the voice bearer is released.

4 Network Impact of Uplink-Triggered VoLTE Mobility

> The feature improves network performance for UEs with voice bearers when fast degradation in uplink radio conditions occurs.

Capacity and Performance

When fast degradation in uplink radio conditions occurs, the feature improves the network performance by performing uplink-triggered intra-LTE blind handover or uplink-triggered measurement-based handover.

Interfaces

No impact.

Other Network Elements

No impact.

5 Parameters for Uplink-Triggered VoLTE Mobility

> This feature introduces MOM attributes for configuration.

Table 2 Parameters

Parameter	Type	Description
EUtranCellFDD.ulVolteCovMobDetect	Introduced	See MOM description.
EUtranCellTDD.ulVolteCovMobDetect	Introduced	See MOM description.
EUtranCellFDD.ulVolteCovMobThr	Introduced	See MOM description.
EUtranCellTDD.ulVolteCovMobThr	Introduced	See MOM description.
ReportConfigA5UlVolte.a5Threshold1Rsrp	Introduced	See MOM description. Note: The configuration of these attributes depends on the network conditions, and the default values are not always recommended. For configuring these attributes, see the other mobility feature descriptions or the RAN Parameter Recommendations Lists.
ReportConfigA5UlVolte.a5Threshold1Rsrq		
ReportConfigA5UlVolte.a5Threshold2Rsrp		
ReportConfigA5UlVolte.a5Threshold2Rsrq		
ReportConfigA5UlVolte.hysteresisA5	Introduced	See MOM description.
ReportConfigA5UlVolte.timeToTriggerA5		
ReportConfigA5UlVolte.triggerQuantityA5		
UeMeasControl.a5TimerUlVolteCovMob		

6 Performance of Uplink-Triggered VoLTE Mobility

> The feature is associated with integrity KPIs and PIs. The feature introduces counters to observe the number of outgoing intra-LTE inter-frequency handovers.

KPIs

Table 3 Key Performance Indicators

KPI	Description
Average DL Packet Error Loss Rate	These KPIs are expected to decrease on the network level. The uplink-triggered VoLTE mobility procedure is performed to avoid service quality loss caused by the fast degradation of uplink radio conditions.
Average DL Packet Error Loss Rate per QCI	
Average UL Packet Loss Rate	
Average UL Packet Loss Rate per QCI	
E-RAB Retainability - Percentage Lost per QCI	This KPI is expected to increase for voice on the network level.
Mobility Success Rate	This KPI is expected to decrease when the value of the `EUtranCellFDD.ulVolteCovMobDetect` or the `EUtranCellTDD.ulVolteCovMobDetect` is set to `BLINDHO`. The decrease might be smaller when the value of the `EUtranCellFDD.ulVolteCovMobDetect` or the `EUtranCellTDD.ulVolteCovMobDetect` attribute is set to `MEASBASEDHO`.
Cell Handover Success Rate	This KPI is expected to decrease when the value of the `EUtranCellFDD.ulVolteCovMobDetect` or the `EUtranCellTDD.ulVolteCovMobDetect` attribute is set to `BLINDHO`. The decrease might be smaller when the value of the `EUtranCellFDD.ulVolteCovMobDetect` or the `EUtranCellTDD.ulVolteCovMobDetect` attribute is set to `MEASBASEDHO`.
Cell Mobility Success Rate	This KPI is expected to decrease when the value of the `EUtranCellFDD.ulVolteCovMobDetect` or the `EUtranCellTDD.ulVolteCovMobDetect` attribute is set to `BLINDHO`. The decrease might be smaller when the value of the `EUtranCellFDD.ulVolteCovMobDetect` or the `EUtranCellTDD.ulVolteCovMobDetect` attribute is set to `MEASBASEDHO`.

Table 4 Performance Indicators

PI	Description
VoLTE UE Satisfaction in Uplink	These PIs are expected to increase. The uplink-triggered VoLTE mobility is performed before the fast degradation in uplink radio condition reduces the service quality.
VoLTE Packet OK Rate	
Differentiated Inter-Frequency Handover Success Rate	
Outgoing Handover Execution SR for each QCI	
Differentiated Handover Execution Success Rate by PLMN and SPID	

The following formulas can be used for monitoring the performance of the feature:

Uplink-Triggered VoLTE Mobility
Differentiated Inter-Frequency Handover
Success Rate:

$$100 \times \frac{\text{pmHoPrepSuccLteInterFUtvm}}{\text{pmHoPrepAttLteInterFUtvm}} \times \frac{\text{pmHoExeSuccLteInterFUtvm}}{\text{pmHoExeAttLteInterFUtvm}} \text{ [\%]}$$

Uplink-Triggered VoLTE Mobility Cell Handover Success Rate:

$$100 \times \frac{\text{pmCellHoPrepSuccLteInterFUtvm}}{\text{pmCellHoPrepAttLteInterFUtvm}} \times \frac{\text{pmCellHoExeSuccLteInterFUtvm}}{\text{pmCellHoExeAttLteInterFUtvm}} \text{ [\%]}$$

Counters

The following PM counters can be used to observe the number of handovers triggered by this feature:

- `EUtranCellFDD.pmCellHoPrepAttLteInterFUtvm`
- `EUtranCellFDD.pmCellHoPrepSuccLteInterFUtvm`
- `EUtranCellFDD.pmCellHoExeAttLteInterFUtvm`
- `EUtranCellFDD.pmCellHoExeSuccLteInterFUtvm`
- `EUtranCellTDD.pmCellHoPrepAttLteInterFUtvm`
- `EUtranCellTDD.pmCellHoPrepSuccLteInterFUtvm`
- `EUtranCellTDD.pmCellHoExeAttLteInterFUtvm`
- `EUtranCellTDD.pmCellHoExeSuccLteInterFUtvm`
- `EUtranCellRelation.pmHoPrepAttLteInterFUtvm`
- `EUtranCellRelation.pmHoPrepSuccLteInterFUtvm`
- `EUtranCellRelation.pmHoExeAttLteInterFUtvm`
- `EUtranCellRelation.pmHoExeSuccLteInterFUtvm`

The following PM counters can be used to observe the number of potential uplink packet loss indication:

- `EUtranCellFDD.pmNumUlPotentialPktLossInd`
- `EUtranCellTDD.pmNumUlPotentialPktLossInd`

Events

The feature has no associated events.

7 Activate Uplink-Triggered VoLTE Mobility

> To use the feature, it must be activated following the usual feature activation procedure.

Prerequisites

- The license key is installed in the node.
- CCTR is active for at least one week before this procedure to collect enough troubleshooting data.
- The Coverage-Triggered Inter-Frequency Handover feature is activated.
- The Intra-LTE Handover feature is activated.

Steps

1. Set the attribute `featureState` to `ACTIVATED` in the `FeatureState=CXC4012603` MO instance.
2. Set the `EUtranCellFDD.ulVolteCovMobDetect` or the `EUtranCellTDD.ulVolteCovMobDetect` attribute to `BLINDHO` or `MEASBASEDHO`.

   If blind handover is chosen, the `EUtranCellFDD.covTriggerdBlindHoAllowed` or the `EUtranCellTDD.covTriggerdBlindHoAllowed` attribute must be set to `true` in the serving cell.

After This Task

Keep CCTR active for at least one week after this procedure for continued collection of troubleshooting data.

8 Deactivate Uplink-Triggered VoLTE Mobility

> If the feature is no longer needed, it can be deactivated following the usual feature deactivation procedure. It must also be deactivated before the activation of any conflicting feature.

Prerequisites

- CCTR is active for at least one week before this procedure to collect enough troubleshooting data.

Steps

1. Set the `EUtranCellFDD.ulVolteCovMobDetect` or the `EUtranCellTDD.ulVolteCovMobDetect` attribute to `DISABLED`.
2. Set the attribute `featureState` to `DEACTIVATED` in the `FeatureState=CXC4012603` MO instance.

After This Task

Keep CCTR active for at least one week after this procedure for continued collection of troubleshooting data.

9 Engineering Guidelines for Uplink-Triggered VoLTE Mobility

> If the Uplink Carrier Aggregation feature is enabled, additional configuration is recommended to avoid affecting the Uplink-Triggered VoLTE Mobility feature.

If the VoLTE Optimized Carrier Aggregation feature is also enabled in addition to the Uplink Carrier Aggregation feature, it is recommended to deconfigure the SCell for the VoLTE session in a carrier aggregation scenario. In this case, UE uplink throughput might be reduced during the VoLTE session because the SCell for uplink carrier aggregation is not available. The SCell can be deconfigured by setting the `EUtranCellFDD.sCellHandlingAtVolteCall` or the `EUtranCellTDD.sCellHandlingAtVolteCall` attribute to one of the following values:

- `DECONF_SCELLS`
- `DECONF_UL_SUPPRESS_DL_SCELLS`
- `DECONF_UL_SCELLS`

If deconfiguring the SCell is not possible, it is recommended to set the same value for the `EUtranCellFDD.ulVolteCovMobThr` or the `EUtranCellTDD.ulVolteCovMobThr` attribute in the PCell and the SCell that contribute to carrier aggregation. This ensures that the handover trigger is more accurate. Triggering the handover can be affected because potential packet loss detection is based on the HARQ response from all cells that participate in uplink carrier aggregation.

Related reference
Dependencies of Uplink-Triggered VoLTE Mobility

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